

Name:

Roll No:

ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI*(An ISO 9001:2015 Certified College)***DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING****B. Tech Degree First Internal Test – March 2023****SIXTH SEMESTER – B. TECH****CST 304: COMPUTER GRAPHICS AND IMAGE PROCESSING**

Time: 2 Hours.

Max.Marks:50

PART - A (Answer all Questions) : 3x5=15 Marks

Qn. No.	Questions	Marks	K level of Qn.	CO & K level
1	Write the flood fill algorithm for filling polygon.	3	K2	CO2(K3)
2	Differentiate between raster scan and random scan display.	3	K1	CO1(K2)
3	What is scan line algorithm?	3	K2	CO2(K3)
4	Calculate the decision parameter for mid-point circle algorithm.	3	K2	CO2(K3)
5	Explain window to viewport transformation with equation.	3	K2	CO3(K3)

PART - B (Answer any 2 Questions) : 2x7=14 Marks

Qn No.	Questions	Marks	K level of Qn.	CO & K level
6	How 8-way symmetry of circle can be used for drawing midpoint circle drawing algorithm, write the symmetric points if (x, y) is a point on the circle with center origin? Explain with an example.	7	K3	CO2(K3)
7	Demonstrate Bresenham's line drawing algorithm. Rasterize the line segment from pixel coordinate (1,1) to (8,5) using Bresenham's line drawing algorithm.	7	K3	CO2(K3)
8	Explain five types of 2D Transformations with relevant equations and diagrams.	7	K2	CO3(K3)

P.T.O

PART C (Answer any 2 Questions) : 2x7=14 Marks

9	Write Mid-point circle algorithm. Use midpoint circle algorithm to plot a circle whose radius =20 units and center is (50,30).	7	K3	CO2(K3)
10	a) Explain the boundary fill algorithm using 4 connected approach.	5	K2	CO2(K3)
	b) Explain how the 4-connected area filling approach differs from 8-connected area filling in boundary filling algorithm.	2		
11	Sketch and explain about Cathode-Ray Tube (CRT) Monitors. Compare Beam-penetration method and Shadow-mask method.	7	K2	CO1(K2)

PART D (Mandatory) : 1x7= 7 Mark

12	A) Translate a square with co-ordinates A(0,0), B(2,0), C(2,2), D(0,2) with translation vector $t_x=2$ and $t_y=3$.	2	K3	CO3(K3)
	B) Perform shear operation on square with co-ordinates A(0,0), B(0,2), C(2,2), D(2,0) in y direction with shearing factor $sh_y=2$.	2		
	C) Write down the equations for 3D rotation with respect to all the coordinate axes. Given a homogeneous point (1, 1, 2), apply rotation 90 degree towards x axis and find out the new coordinate points.	3	K3	CO3(K3)

Name.....

Roll No.....

ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI*(An ISO 9001:2015 Certified College)***B. TECH DEGREE SECOND INTERNAL EXAM – JUNE 2023****DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING****SIXTH SEMESTER – B. TECH A&B****CST 304: COMPUTER GRAPHICS AND IMAGE PROCESSING****PART - A (Answer all Questions): 3x5=15 Marks**

Qn. No.	Question	Marks	K level	CO and K level
1	Describe the concept of vanishing point with relevant figures.	3	K2	CO4 & K2
2	What is scan line algorithm?	3	K1	CO4 & K2
3	Explain how to measure the distance between two pixels?	3	K1	CO5 & K3
4	Define the terms with respect to pixels: (i) Adjacency (ii) Resolution	3	K1	CO5 & K3
5	Describe the basic gray level transformations with graph.	3	K2	CO6 & K3

PART - B (Answer any 2 Questions): 2x7=14 Marks

6	Use Cohen-Sutherland Algorithm to clip the line P1(70,20) and P2(100,10) against a window A(50,10), B(80,10), C(50,40) & D(80,40).	7	K3	CO3(K3)
7	Write note on Parallel and Perspective Projections.	7	K2	CO3(K3)
8	a) Explain Z Buffer Algorithm for hidden surface elimination. b) With diagram, explain the 3D viewing pipeline	4 3	K1	CO4(K2)

PART C (Answer any 2 Questions): 2x7=14 Marks

9	a) What are the fundamental steps in image processing? b) Explain the components of image processing	4 3	K1	CO5(K3)
10	Consider an image segment shown below:		K3	CO5(K3)

	3 1 2 1(q) 2 2 0 2 1 2 1 1 (p)1 0 1 2 Let $V = \{1, 2\}$ and compute the length of shortest 4-, 8- and m- path between p and q. If a particular path does not exist between these two points, explain why?	7		
11	a) Explain in brief the spatial convolution operation. b) What are the processes involved in converting an image into digital image.	4 3	K1	CO5(K3)
PART D (Mandatory): 1x7= 7 Mark				
12	Perform histogram equalization of the image $\begin{bmatrix} 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \\ 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 5 & 4 & 3 \\ 4 & 4 & 4 & 4 & 4 \end{bmatrix}$	7	K3	CO6(K3)

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**B.Tech Degree S6 (R, S) / S4 (PT) (R, S) Examination June 2023 (2019 Scheme)****Course Code: CST304****Course Name: COMPUTER GRAPHICS AND IMAGE PROCESSING**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

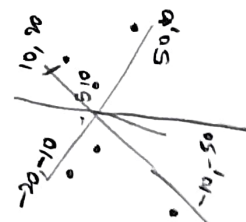
- | | | |
|----|--|-----|
| 1 | ✓ Differentiate the aspect ratio and resolution of a raster scan display? | (3) |
| 2 | List out the applications of computer graphics. | (3) |
| 3 | Write down the 4- neighbour Flood-filling algorithm | (3) |
| 4 | ✓ Describe the basic 3-Dimension transformations | (3) |
| 5 | ✓ Discuss the steps involved in window to viewport coordinate transformation in 2D? | (3) |
| 6 | ✓ Illustrate the Oblique Projections | (3) |
| 7 | ✓ Define the terms related to pixel of an image:
i) Neighbours ii) Boundary iii) Path | (3) |
| 8 | ✓ Differentiate gray scale image and a colour image | (3) |
| 9 | ✓ Write the algorithm for basic Global thresholding | (3) |
| 10 | ✓ Compare Smoothing and sharpening techniques in image processing | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

- | | | |
|----|---|-----|
| 11 | a) Explain the architecture of raster scan system with suitable diagrams. | (6) |
| | b) Rasterize the line segment from pixel coordinate (1, 1) to (8, 5) using Bresenham's line drawing algorithm | (8) |

OR

- | | | |
|----|---|------|
| 12 | a) Scan convert the line segment with end points (0,0) and (10,5) using DDA line drawing algorithm. Find out and discuss the advantages and disadvantages of this method. | (10) |
|----|---|------|



- b) Describe the working of a beam penetration CRT (4)

Module II

- 13 a) Compare the purpose of flood-fill and boundary-fill algorithm. Discuss the data structures used for scan-line polygon filling algorithm. (8)
- b) Given a triangle A(20,10) B(80,20) C(50,70). Find the co-ordinates of vertices when (a) Reflection about the line $x=y$, (b) Reflection about the diagonal $y = -x$. (6)

OR

- 14 a) Consider a triangle at (2,2), (10,2), (2,10). Perform the following 2D transformations in succession and find the resultant vertices (10)
- (i) Scale with respect to (2,2) by scaling factors (2,2) respectively along x and y directions.
- (ii) Rotate by 90° counter clockwise direction
- b) Discuss the techniques used to identify the inside and outside points of a polygon. (4)

Module III

- 15 a) Write Cohen Sutherland Algorithm and illustrate the region and region code (8)
- b) Summarize on multi view and axonometric projections. Write the equation for projection coordinates of a point P(x,y,z), if the view plane is placed along z-axis. (6)

OR

- 16 a) Describe the Depth buffer algorithm used for visible surface detection (8)
- b) List out the applications of visible surface detection algorithms (6)

Module IV

- 17 a) Write short notes on (i) Illumination, (ii) Reflectance (6)
- b) Describe the components of an image processing system with necessary diagram (8)

OR

- 18 a) Define convolution. List out the properties of convolution. Illustrate the steps involved in convolving an image with a mask. (8)
- b) List out the applications of image processing in Medical field (6)

Module V

- 19 a) A 3-bit image of size 4×5 is shown below. Compute the histogram equalized (8)

image.

0	1	1	3	4
7	2	5	5	7
6	3	2	1	1
1	4	4	2	1

- b) Define edges in an image. Discuss on any two edge filters (6)

OR

- 20 a) Discuss on region based approaches used for segmentation (8)
b) Describe the Laplacian filter masks used for image processing (6)
